

Every sufficiently large quadratic size is possible for one die

Research corollary; two independent full reviews completed

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Theorem 1. *There is an absolute constant C such that, for every $n \geq 2$ and every integer $K \geq Cn^2$, there is a tie-free permutation-fair collection of n equiprobable finite dice in which a specified die has exactly K faces. The other $n - 1$ dice can all have the same number of faces, which is not bounded quantitatively here.*

This strengthens the reviewed existence of some exceptional size $K = O(n^2)$. In conjunction with the reviewed universal lower bound $K \geq cn^2$, it locates the entire possible-size spectrum up to absolute constant factors. No divisibility condition is imposed on K : denominator clearing occurs only on the other dice.

Lemma 1. *For every $m \geq 1$ and every integer $K \geq C_0(m + 1)^2$, there is a real equal-weight quadrature for the uniform probability measure on $[0, 1]$, exact through degree m , with exactly K distinct nodes strictly inside $(0, 1)$.*

Proof. Gilboa–Peled, *Chebyshev-type Quadratures for Doubling Weights*, Theorem 1.1, bounds both $N_w(d)$ and $\overline{N}_w(d)$ by a constant multiple of their local-mass quantity $R_w(d)$. Their definition of $\overline{N}_w(d)$, on pages 2–3, requires a rule with exactly N nodes for every $N \geq \overline{N}_w(d)$. For the constant weight on $[-1, 1]$, the smallest window mass is of order d^{-2} ; hence $\overline{N}_w(d) \leq C_1 d^2$. Affine rescaling gives the same statement on $[0, 1]$. The primary text and theorem are available at <https://arxiv.org/pdf/1507.01505>, pages 2–3.

Apply that result at degree $2m + 2$, choosing the prescribed K . Increase C_0 so that it also ensures $K \geq m^2$. The resulting rule has at least $m + 2$ distinct support points: if it had at most $m + 1$, the square of the support-vanishing polynomial would have degree at most $2m + 2$, zero quadrature sum, and strictly positive integral. Thus at least m distinct support points are interior.

Select one node at each of m distinct interior points as pivots. The Jacobian of the first m moment sums in these coordinates is an invertible scaled Vandermonde matrix. By the implicit function theorem, small arbitrary perturbations of the remaining coordinates can be compensated by the pivots while preserving all these moments. Move free endpoint coordinates slightly inward. Pivot coordinates remain interior and distinct. Free–free collisions are proper hyperplanes. At any initial collision of a free node and a pivot, the matching pivot has derivative -1 when that free node varies, so their difference has nonzero derivative -2 . These collisions are therefore proper analytic zero sets. Initially absent collisions remain absent after taking a small neighborhood. A finite union of proper analytic zero sets does not fill the open set of permitted inward perturbations. A collision-free choice, followed by sorting, gives exactly K distinct interior nodes. $\square$

Proof of the theorem. Set $m = n - 1$ and use the preceding lemma at the prescribed K . The reviewed proof in `../..size_bounds/polynomial_exceptional_die.tex` now applies without repeating, adding, or removing any distinguished nodes. We detail why its quantifiers permit this application.

That proof requires a degree- m equal-weight real rule with $K \geq m^2$ distinct interior nodes, and no upper bound on K . It selects disjoint sets $Q_1, \dots, Q_m$, each containing m of those nodes,

and rational moment bumps with disjoint supports around the nodes. The node locations are approximated by rationals. For each old die, an invertible m -by- m Vandermonde system finds its correction coefficients. These coefficients tend to zero as the rational approximation approaches the real rule, so all corrected densities remain positive. The disjoint-support response identity then makes every full order have probability $1/n!$.

The distinguished distribution is still uniform on the original number K of rational locations. All other distributions have positive rational piecewise-constant densities. Partitioning at their breakpoints and at the K distinguished locations, then replacing each interval by a finite fair block, preserves the complete order law. A common denominator D for the interval masses and a common F -face fair block give exactly FD faces for each old die. One distinguished face is inserted at each of the K selected locations. Assigning globally distinct integer labels finishes the construction with exactly the requested distinguished size. $\square$

The finite realization may require very large F and D . This corollary gives neither a small total-size construction nor simultaneous control of the sizes of several specified dice.